

2015

Operating Systems*Paper - 302**Full Marks - 70**Time - 3 hours**Answer any five questions.**The figures in the margin indicate full marks.*

Candidates are required to give their answers in their own words as far as practicable

Illustrate the answer whenever necessary

1. (a) Distinguish between the following:
 - i. Multiprogramming and Multiprocessing.
 - ii. Network system and Distributed system.
- (b) What is a real time system?
- (c) What do you mean by context switching?
- (d) What are the various states of a process? Show them by means of the state diagram and explain the different state transitions.

 $(2 + 2) + 2 + 2 + (2 + 4)$

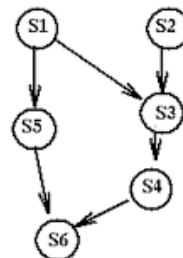
2. (a) Distinguish between preemptive and nonpreemptive CPU scheduling.
- (b) Using the following information, draw the Gantt charts and find out the average waiting time, average response time and average turnaround time of the five processes for the following CPU scheduling algorithms: i) Shortest Remaining Time First and ii) Round Robin with time slice 3.

Process	Arrival time	Burst time
P1	0	4
P2	2	3
P3	6	7
P4	8	1
P5	12	8

- (c) When does Round Robin scheduling behave like FCFS scheduling?
- (d) Explain the *exponential averaging* method for estimating the length of the next CPU burst.

 $2 + (2 \times 3) + 2 + 4$

3. (a) "Context switching among threads is less costlier than that among processes"-Justify.
- (b) Discuss the differences between user level threads and kernel level threads.
- (c) Describe the various multi-threading models.
- (d) Convert the following precedence graph into a program using Fork/Join construct.

 $3 + 4 + 3 + 4$

4. (a) What is meant by critical section of a process?
- (b) What are the necessary conditions to achieve mutual exclusion?
- (c) Define the *Wait* and *Signal* operations on semaphore that do not involve busy waiting.

- (d) Discuss the solution to the Reader-Writer problem using semaphore and write down the pseudo codes for the reader and writer processes. processes.

$$2 + 3 + (2 + 2) + 5$$

5. (a) What are the necessary conditions for a deadlock to occur?
 (b) Does a cycle in the resource allocation graph always indicate a deadlock? Justify your answer.
 (c) Consider the following system with 5 processes, P0 through P4 and four resource types, A, B, C, and D. At time T0, the following snapshot of the system was taken:

Process	Allocation				Max			
	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2
P1	1	0	0	0	1	7	5	0
P2	1	3	5	4	2	3	5	6
P3	0	6	3	2	0	6	5	2
P4	0	0	1	4	0	6	5	6

If the available number of instances for resource types A, B, C and D are 1, 5, 2 and 0, respectively, then answer the following:

- What is the content of the matrix *Need*, which stores the number of resources still needed by each process?
- Determine whether the system is in *safe state* or not using *banker's algorithm*. If it is in safe state, then give a safe sequence of the processes.

$$4 + 3 + (2 + 5)$$

6. (a) Briefly describe the hierarchy of memory organization.
 (b) What do you mean by internal and external fragmentation? Give examples.
 (c) Consider a pure paging system along with TLB for storing the most recent part of the page map table. If TLB hit ratio is 85%, main memory access time is 50 nanoseconds and TLB access time is 10 nanoseconds, what will be the overall average access time?
 (d) What are the factors that affect page sizes in a pure paging system?
 (e) What is sharing easier in segmentation than paging?

$$3 + 3 + 2 + 4 + 2$$

7. (a) Consider the following page reference string:

1 2 3 2 1 5 2 1 6 2 5 6 3 1 3 6 1 2 4 3

If the number of free frames in memory is 3 then find the number of page faults for the following page replacement strategies: i) FIFO, ii) Least Recently Used.

- (b) Discuss the *working set principle* for handling the thrashing problem. What is the disadvantage of working set model?
 (c) Consider a two dimensional array A[100][100], each element taking 2 bytes. Suppose page size is 50 bytes. Page replacement strategy is FIFO and the number of available frames is 1. How many page faults will occur to initialize the matrix A in i) row wise and ii) column wise?

$$(2 + 3) + (3 + 2) + (2 + 2)$$

8. (a) Consider the following sequence of disc requests (track numbers): 45, 80, 64, 19, 120, 12, 62, 93. All the requests have come at the same time. The initial head position is on track 70. The range of tracks is from 0 to 199 and the disc head needs 1 ns to move between two consecutive tracks. Find out the total time to serve all the requests for the following disc scheduling algorithms: i) SSTF, ii) LOOK.
 (b) Write short notes on the following (*any two*):
 i. Process Control Block.
 ii. Deadlock prevention.
 iii. Multi-level page map table.
 iv. Clock algorithm for page replacement.

$$(3 + 3) + (2 \times 4)$$

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